AHMED ALI KHAN

AWS CLOUD AND DEVOPS ENGINEER

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PROFESSIONAL SUMMARY

AWS Cloud and DevOps Engineer with nearly 5 years across cloud engineering and support. Builds multi-account AWS foundations and production EKS platforms, replacing manual operations with Terraform and secure delivery workflows. Owns infrastructure decisions through implementation, cost optimization and incident response.

TECHNICAL SKILLS

AWS: Organizations, Control Tower, EC2, S3, RDS, Lambda, ECR, DataSync, IAM Identity Center
Infrastructure as Code: Terraform, TFLint, Checkov
Kubernetes: Amazon EKS, Docker, Helm, Karpenter, HPA, RBAC, IRSA, Kyverno, External Secrets Operator
CI/CD: GitHub Actions, Actions Runner Controller (ARC), Argo CD, GitOps, OIDC
Networking: VPC, Transit Gateway, AWS RAM, Route 53, ALB, PrivateLink, AWS Network Firewall
Security: IAM, SCPs, KMS, Secrets Manager, Wiz, Prisma Cloud, GitHub Advanced Security (GHAS), Trivy
Observability: CloudWatch, Datadog, Dynatrace, Prometheus, Grafana
Automation: Linux, Python, Boto3, Bash, Git, AWS Systems Manager
Data Platforms: Databricks on AWS, Unity Catalog, Delta Lake
Governance: AWS Config, CloudTrail, NIST SP 800-53, CIS Benchmarks

PROFESSIONAL EXPERIENCE

Franklin Infotech | Ann Arbor, MI **Feb 2024 - Mar 2026**
Cloud Platform Engineer

- Cut EKS compute spend 22% (~$48,000 annualized) by rightsizing requests and using Karpenter consolidation. Reserved On-Demand nodes for critical services and Spot for interruptible workloads.
- Closed 24 cloud security and workload-access findings by correcting IAM permissions and S3 exposure. Prioritized remediation with Wiz and codified preventive controls in Terraform.
- Built and optimized AWS landing zones with Organizations and Control Tower. Separated production accounts, centralized CloudTrail logging and enforced SCP guardrails with IAM Identity Center access.
- Brought console-managed resources into Terraform without replacing live infrastructure. Separated remote state by account and environment, refactored modules and enforced TFLint, Checkov and drift review.
- Built EKS clusters and led staged upgrades after validating deprecated APIs and add-on compatibility. Rotated nodes with spare capacity and Pod Disruption Budgets to protect application availability.
- Recovered EKS workloads from pod scheduling, VPC CNI address exhaustion and DNS failures. Used Kubernetes events and Datadog to isolate resource constraints and prevent repeat incidents.
- Made releases reproducible with GitHub Actions, Helm and Argo CD. Promoted the same ECR image digest across environments, using health checks and reviewed Git reverts for recovery.
- Secured AWS infrastructure deployments with repository- and environment-scoped OIDC trust. Required CodeQL SAST, dependency review and Trivy image checks before merge, with secret-scanning push protection.
- Replaced persistent build hosts with ephemeral ARC runners in an isolated EKS build cluster. Limited runner groups to approved repositories and separated build permissions from production deployment roles.
- Deployed Databricks workspaces in customer-managed AWS VPCs with Terraform. Governed bronze/silver/gold Delta Lake access through Unity Catalog and scoped S3 storage credentials, restricting direct access that bypassed catalog controls.
- Restricted AI workload access through IRSA roles and IAM permission boundaries. Scoped Bedrock invocation and S3/KMS permissions, using VPC endpoints and endpoint policies for private service access.
- Blocked privileged pods and unapproved registries with Kyverno. Enforced non-root execution and restricted Linux capabilities after testing policies in audit mode and resolving application exceptions.
- Removed secrets from Helm values with External Secrets Operator and Secrets Manager. Used namespace-specific service accounts with scoped IRSA roles and tested application reloads after credential rotation.

CEO Foundry | Ypsilanti, MI **May 2023 - Feb 2024**

DevOps Engineer

- Removed long-lived application credentials from code and deployment workflows by adopting Secrets Manager and scoped IAM roles. Separated KMS key administration from application decryption permissions.
- Migrated Azure Blob Storage data to S3 using AWS DataSync. Reconciled object inventories and completed a final incremental sync during a controlled write pause before application cutover.
- Standardized development, test and staging with reusable Terraform modules and GitHub Actions approvals. Isolated environment state and deployment roles to prevent changes from crossing account boundaries.
- Connected application VPCs through a Transit Gateway shared with AWS RAM. Isolated environments through route-table associations and propagation, allowing only approved paths to shared services.
- Remediated S3 exposure and excessive permissions with Prisma Cloud and AWS Config. Mapped encryption, access restrictions and CloudTrail evidence to selected NIST SP 800-53 controls.
- Improved incident diagnosis with CloudWatch alarms and Datadog dashboards for application errors and host saturation. Added recovery runbooks and resolved recurring issues through root-cause reviews.
- Reduced idle nonproduction spend with scheduled EC2 stop/start automation through EventBridge and Lambda. Added ownership tags and AWS Budgets alerts so teams could act on unexpected usage.

Proximate Technologies | Ypsilanti, MI **Nov 2021 - May 2023**

Cloud Engineer

- Saved approximately 25 hours of manual work monthly by automating recurring infrastructure and data-processing tasks with Python, Boto3 and event-driven jobs, including failure logging for operational follow-up.
- Designed AWS reference architectures for 20+ application and data workloads. Documented service trade-offs, network boundaries and recovery requirements so teams could review operational risks before implementation.
- Standardized networking and IAM across three AWS accounts using Organizations and reusable Terraform modules. Replaced ad hoc console changes with reviewed GitHub Actions workflows.
- Improved EC2 queue-processing reliability by tuning Auto Scaling to workload demand and configuring health checks. Used Datadog to correlate application failures with host metrics during incidents.
- Validated disaster recovery by rebuilding infrastructure with Terraform and restoring RDS backups in isolation. Tested application connectivity and permissions, recording recovery times and correcting runbook gaps.
- Automated RDS credential rotation with Secrets Manager and Lambda. Tested new database credentials and application reconnection before completing rotation, replacing manual password updates.
- Restricted traffic with Terraform-managed security groups and AWS Network Firewall. Verified symmetric routes through firewall endpoints and used VPC Flow Logs to troubleshoot blocked connections.
- Reduced unnecessary S3 storage costs with lifecycle transitions matched to access patterns and retention needs. Applied versioning and selective cross-Region replication to datasets with regional recovery requirements.
- Automated EC2 patching with Systems Manager maintenance windows and post-patch health checks. Used Session Manager for administrative access, removing the need for inbound SSH on managed instances.

Smart Bridge Educational Services | Vijayawada, India **Aug 2017 - Mar 2018**

Cloud Support Engineer Intern

- Improved EC2 release consistency with Bash checks for Java and Spring Boot configuration. Investigated failures through CloudWatch logs and Linux diagnostics, documenting recovery steps for engineers.

CERTIFICATIONS AND ACCREDITATIONS

Certified Kubernetes Administrator (CKA) | AWS Certified Solutions Architect - Associate
HashiCorp Certified: Terraform Associate | Databricks AWS Platform Architect Accreditation

EDUCATION

Illinois Institute of Technology | Chicago, IL May 2021
M.S. Information Technology and Management | GPA 3.8

JNTU Kakinada | Andhra Pradesh, India 2019
B.Tech. Electronics and Communication Engineering